

The Token Detail Record (TDR)

A transaction record for AI consumption, attribution, rating, billing and reconciliation

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Publication statement

This paper presents and formally defines the term Token Detail Record (TDR) as a record type for an individual AI inference transaction. The term is used here for the metering, attribution, rating, billing, audit and reconciliation of AI consumption, with particular attention to systems that route requests across models and providers.

Scope of disclosure

This paper describes the purpose and information classes of a TDR. It does not publish the complete production field layout, ordinal positions, transport rules, rating formulas or archive mechanics of the related implementation specification.

Abstract

AI services are commonly billed from provider-reported usage totals. That is sufficient for a simple application using one model and one vendor. It becomes less satisfactory when a single business action can invoke several models, pass through routing logic, reuse cached context, generate retries, or create a chain of machine-originated requests. The underlying commercial event is no longer well described by an invoice total or a provider dashboard.

The Token Detail Record, or TDR, is proposed as the atomic accounting record for that environment. One TDR represents one completed inference round trip and preserves the information needed to identify the event, attribute it, understand how it was routed, capture usage, associate rates and charges, distinguish reported quantities from independent measurements and estimates, and correlate the event with a larger interaction.

The idea follows a long-standing lesson from telecommunications: high-volume usage can only be audited reliably when the aggregate bill can be traced back to durable transaction records. The TDR applies that principle to model consumption rather than call duration.

1. Definition

Token Detail Record (TDR). A durable transaction record describing one completed AI inference event, including the information required to identify, attribute, measure, rate, correlate and audit that event.

A TDR is not a prompt log and it is not merely a token counter. Its purpose is accounting and operational evidence. The record should remain useful after price tables, model aliases, routing policies and provider dashboards have changed.

2. Why an atomic record is needed

An invoice is an aggregate. A dashboard is a presentation. Neither is a substitute for the underlying transaction population when a customer needs to answer questions such as: Which application incurred this cost? Which provider served the request? Was the requested model the model that actually ran? How much of the input came from cache? What rate was in force? Was a failure retried? Did one user action create ten downstream model calls?

Those questions require evidence at the event level. A TDR supplies that evidence in a form that can be retained, summed and reconciled.

Core design principles

- One record represents one completed inference round trip, not separate prompt and response accounting fragments.
- The record separates identity, attribution, routing, quantities, commercial terms and correlation rather than collapsing them into a single usage number.
- Reported, measured and estimated quantities are distinguishable.
- The accounting record can be retained without storing prompt or completion content.

- Records can be related to one another so a multi-step agent interaction can be reconstructed as a transaction tree.

3. Information classes within a TDR

A production TDR contains multiple critical fields. The precise layout is implementation-specific, but the information falls into a stable set of sections. The examples below are illustrative rather than a complete schema.

Section	Why it matters	Illustrative fields
Record identification	Establishes a unique event, when it occurred, and the accounting period to which it belongs. This is the basis for duplicate control, replay analysis and audit traceability.	request identifier; event timestamp
Attribution	Answers who or what caused the consumption and where the expense belongs. Attribution supports customer billing, departmental chargeback, application economics and project accounting.	client or application identifier; project or cost-centre tag
Routing and disposition	Records where the request went, what model actually served it, and how the event ended. This matters when a routing layer can substitute models, use fallback providers or terminate a request before a normal response.	provider; model used; terminal status
Performance	Provides service-quality evidence alongside cost. A transaction can be correctly billed yet operationally unacceptable.	end-to-end latency; intermediary overhead
Provider-reported quantities	Preserves the usage quantities supplied by the upstream model provider. These values are important because provider billing practices are not uniform and may distinguish ordinary input, cached input, output and other token classes.	input tokens; output tokens; cache-read tokens
Rating inputs	Records the commercial terms in force at the time of the event so the event can be re-rated later without relying on a current price page.	rate-card version; applicable input or output rate
Charges	Carries the monetary result associated with the event. Event-level charges allow a bill to be reconstructed from its underlying population rather than accepted only as an aggregate.	input charge; output charge
Independent measurements	Captures observations made by the intermediary itself rather than simply repeating provider accounting. Independent measurements are useful for reasonableness checks, anomaly detection and reconciliation research.	characters submitted; characters returned
Estimated decomposition	Preserves useful derived quantities that a provider may not expose directly. Estimates must be marked as estimates so they cannot be mistaken for measurements or provider-reported usage.	estimated current-turn usage; estimated carried-context usage
Integrity references	Provides a way to associate the accounting record with protected evidence without placing the underlying content in the TDR.	payload digest or equivalent integrity reference

Diagnostics	Supplies enough operational context to classify failures while keeping the record focused on accounting rather than application logging.	application version; numeric error class
Summation and correlation	Connects one inference event to a larger business interaction. This becomes essential when agents call agents, retries occur, or one user request creates a tree of model activity.	root interaction identifier; parent event identifier; acting-agent identifier

4. Three sources of truth must remain distinct

A useful TDR should not blur all quantities into one undifferentiated set. There are at least three materially different sources of information.

Provider-reported: Values returned by the model provider in its response or usage envelope. These are commercially important because they often form the provider's own billing basis.

Independently measured: Values observed by the proxy, gateway or other intermediary. These provide a second evidentiary source and can be compared with provider-reported quantities.

Derived or estimated: Values calculated because a useful decomposition is not directly reported. These can support analysis, but they should carry their own provenance or estimator version and must not be presented as measurements.

This distinction matters for reconciliation. The party issuing an invoice should not be the only possible source of evidence used to examine that invoice. A TDR can preserve the provider's account of usage while also retaining independent observations and clearly labelled estimates.

5. TDRs in agentic systems

The need for correlation grows as AI systems become more autonomous. A user may initiate one business action, but the application may ask an orchestrator to plan the work, call a research model, invoke a specialist, retry a failed request and ask another model to review the result. Commercially, that is one business interaction. Operationally, it is a population of inference events.

A TDR should therefore support correlation beyond a flat transaction identifier. Root, parent and actor relationships allow individual inference events to be reconstructed into a tree. That makes it possible to calculate the true model cost of a workflow and to distinguish productive work from retry loops, fallback activity or unnecessary context carry-forward.

6. Content boundaries

An accounting record does not need to become a second repository for model content. A TDR can exclude prompt text, completion text, provider error bodies and credential material while preserving the accounting evidence needed for audit and reconciliation. Where evidence must be associated with protected payload material, an integrity reference can be retained instead.

This separation is important for privacy, security and retention policy. Accounting data often needs to be retained longer and accessed by different teams than the underlying conversational content. Keeping the two purposes separate reduces the amount of sensitive material that must travel with billing data.

7. Relationship to a detailed implementation specification

The TDR concept should be broader than any one physical record layout. A production implementation may contain dozens of individual elements, precise data types, field ordering rules, rating precision, duplicate-handling rules, archive conventions and delivery guarantees. Those details belong in an implementation or interface specification.

The public concept is simpler: a TDR is the durable, event-level record from which AI usage can be identified, attributed, rated, correlated, summed and audited. That definition allows implementations to evolve without changing the underlying accounting concept.

8. Relationship to telecommunications accounting

The term is intentionally analogous to the Call Detail Record used in telecommunications. A CDR made high-volume calling activity auditable by reducing each chargeable event to a durable record that could be mediated, rated, billed and reconciled. AI inference has a different unit of consumption and a more varied set of providers, models and usage categories, but the accounting problem is familiar.

A TDR applies the same discipline to token-based and model-based consumption. It is not a claim that token usage behaves like call duration. It is a claim that high-volume machine transactions need an atomic record if their aggregate charges are to be explainable later.

9. Proposed industry use

The term Token Detail Record should be useful independent of any particular vendor. An enterprise, model provider, gateway, billing platform or audit system can use the term for an event-level AI accounting record, provided the record preserves enough information to support identification, attribution, usage measurement, commercial rating and reconciliation.

Standard terminology is valuable because the market currently uses several different artifacts for parts of the same problem: API logs, usage reports, cost dashboards, traces and invoice exports. A TDR names the accounting object itself.

10. Origin and publication statement

Jerry Lambert developed and uses the term Token Detail Record (TDR) for the event-level accounting record described in this paper and in the Foundation Specification. This publication is intended to establish a dated public definition of the term and its use in AI consumption accounting.

The historical claim made here is specific: this publication documents the author's introduction and formal definition of TDR for AI inference metering, attribution, rating, billing and reconciliation. It does not attempt to prove that the three words "token detail record" have never appeared previously in any unrelated context.

Foundation Specification

AGU-TDR-001, Token Detail Record - Mediation Output Specification, Issue 1, August 2026, Record Format Version 1.7. This document serves as the Foundation Specification for the TDR concept described here. It provides a detailed working implementation covering record generation, field layout, conformance, delivery and exception handling. [View the Foundation Specification](#)

Canonical short definition

A Token Detail Record (TDR) is a durable transaction record describing one completed AI inference event and preserving the information required to identify, attribute, measure, rate, correlate and audit that event.